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[23BSACH003]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

APPLIED CHEMISTRY

Time: 3 Hours Maximum **Marks: 100**

Instructions:
i. Missing data may be suitably assumed.
ii. Answer All Questions

		PART - A			**05 x 02=10**
1.	**a.**	What is an electrochemical cell?	L1	CO1	**02**
	b.	Define two properties of system with suitable examples.	L2	CO2	**02**
	c.	Describe the Pauli's exclusion principle.	L2	CO3	**02**
	d.	What is meant by Positional isomerism?	L1	CO4	**02**
	e.	What is the word Nano refer to? Who coined the term, from which language and write the meaning?	L1	CO5	**02**
		PART - B			**06 × 05=30**
2.	**a.**	Differentiate between Galvanic cell and Electrolytic cell.	L2	CO1	**05**
		OR			
	b.	The standard electrode potential (E°) for Daniel cell is +1.1 V. Calculate the ΔG° for the reaction $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ (1 F = 96500 C mol^{-1}).	L3	CO1	**05**
3.	**a.**	Explain the characteristics of Ellingham diagram.	L2	CO2	**05**
		OR			
	b.	Differentiate between exothermic and endothermic reaction with neat diagram.	L2	CO2	**05**
4.	**a.**	Summarize the concept of effective nuclear charge.	L3	CO3	**05**
		OR			
	b.	What is bond order? Draw the molecular orbital diagram and calculate the bond order for H_2?	L1	CO3	**05**
5.	**a.**	Elaborate the chemical structure and synthesis of aspirin molecule.	L3	CO4	**05**
		OR			
	b.	What are nucleophilic substitution reactions? Give any two examples?	L1	CO4	**05**
6.	**a.**	Write a short note on ball milling method of synthesis of nanoparticles.	L2	CO5	**05**
		OR			
	b.	Interpret the instrumentation of scanning electron microscope.	L3	CO5	**05**
7.	**a.**	Draw the molecular orbital diagram and calculate the bond order of HF.	L3	CO3	**05**

		OR			
	b.	Describe the types of elimination reaction with examples.	L2	CO4	**05**
		PART - C			**06 × 10=60**
8.	**a.**	Discuss the following: **i.** Differential metal corrosion and **ii.** Differential aeration corrosion.	L2	CO1	**10**
		OR			
	b.	Elaborate the construction and working of Primary reference electrode? Give the limitations of Standard Hydrogen Electrode.	L3	CO1	**10**
9.	**a.**	Explain the crystal filed splitting in tetrahedral complex with neat diagram.	L2	CO2	**10**
		OR			
	b.	Summarize the factors affecting equilibrium constant and application of equilibrium constant.	L3	CO2	**10**
10.	**a.**	Draw the MO energy level diagram for O_2 and N_2. Show which orbital occupied and workout the bond orders of these molecules.	L3	CO3	**10**
		OR			
	b.	What is electro negativity and electron affinity? Explain its trend in periodic table in detail?	L2	CO3	**10**
11.	**a.**	Describe the structural isomerism? Explain the different types of structural isomerism with the help of examples.	L2	CO4	**10**
		OR			
	b.	Explain the rules for assigning R and S configuration.	L3	CO4	**10**
12.	**a.**	Interpret the principle, instrumentation and applications of FTIR spectroscopy.	L3	CO5	**10**
		OR			
	b.	Write the instrumentation and applications of UV-VIS Spectrophotometer.	L1	CO5	**10**
13.	**a.**	Draw the shapes of the d-orbital and explain why they are split into two groups t_{2g} and e_g in octahedral ligand field.	L3	CO3	**10**
		OR			
	b.	Illustrate the synthesis of metal oxide nanoparticles using sol-gel method.	L3	CO5	**10**